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Small air-cooled engine assembly with dry sump lubrication system

Abstract

An engine assembly includes a small air-cooled engine and a dry sump lubrication system including an external oil reservoir, wherein the dry sump lubrication system has an overall oil capacity that provides at least five hundred hours of engine oil life.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 15/775,704, filed on May 11, 2018, which claims priority to PCT Application No. PCT/US2016/058,909, filed on Oct. 26, 2016, which claims the benefit U.S. Provisional Patent Application No. 62/335,500, filed May 12, 2016, and U.S. Provisional Patent Application No. 62/255,227, filed Nov. 13, 2015, and is a continuation of U.S. patent application Ser. No. 15/295,294, filed Oct. 17, 2016, all of which are hereby incorporated by reference in their entireties.

BACKGROUND
(1) The present invention relates generally to the fields of small internal combustion engines and outdoor power equipment, and in particular, engine assemblies including small internal combustion engines and dry sump lubrication systems, external oil reservoirs for use with dry sump lubrication systems, oil filters for use with dry sump lubrication systems, and outdoor power equipment including dry sump lubrication systems.
SUMMARY
(2) One embodiment of the invention relates to an engine assembly including a small air-cooled engine and a dry sump lubrication system including an external oil reservoir, wherein the dry sump lubrication system has an overall oil capacity that provides at least five hundred hours of engine oil life.
(3) In some embodiments, the external oil reservoir has an oil fill capacity of at least five quarts.
(4) In some embodiments, the dry sump lubrication system includes a crankcase chamber; wherein an intended oil fill level of the crankcase chamber is between 2.5 quarts to 3 quarts.
(5) In some embodiments, the small air-cooled engine includes a crankshaft including a counterweight positioned in a crankcase chamber. In some embodiments, a recommended oil level of the crankcase chamber during normal operation of the engine is below a lowermost portion of the counterweight.
(6) In some embodiments, an intended oil fill level of the crankcase chamber is between 2.5 quarts to 3 quarts.
(7) In some embodiments, the small air-cooled engine includes an engine block, an oil gallery configured to distribute oil, a crankcase cover, and a crankcase chamber defined by the engine block and the crankcase cover. The dry sump lubrication system includes the external oil reservoir, a supply pump, and a return pump. The external oil reservoir includes an oil tank defining an oil chamber and an oil filter assembly. The oil filter assembly includes a housing and a filter. The housing at least partially defines a filter chamber. The filter chamber is in fluid communication with the oil chamber. The filter is positioned within the filter chamber. The supply pump is in fluid communication with the oil chamber and the oil gallery. The supply pump is configured to draw oil from the oil chamber and provide pressurized oil to the oil gallery. The return pump is in fluid communication with the crankcase chamber and the filter chamber. The return pump is configured to draw oil from the crankcase chamber and provide pressurized oil to the filter chamber. The filter is configured to filter the pressurized oil provided to the filter chamber.
(8) In some embodiments, the return pump is configured to pressurize oil to less than ten pounds per square inch.
(9) In some embodiments, the crankcase chamber and the oil chamber have an overall oil capacity that provides at least five hundred hours of engine oil life between routine oil changes.
(10) In some embodiments, the crankcase chamber has a crankcase volume and the oil chamber has

a reservoir volume greater than the crankcase volume.
(11) In some embodiments, the reservoir volume is at least five quarts.
(12) In some embodiments, an intended oil fill level of the crankcase chamber is between 2.5 quarts to 3 quarts.
(13) In some embodiments, the reservoir volume is greater than five quarts and a recommended oil fill capacity of the oil chamber is five quarts.
(14) In some embodiments, the engine assembly includes a crankshaft including a counterweight positioned in the crankcase chamber. In some embodiments, a recommended oil level of the crankcase chamber during normal operation of the engine is below a lowermost portion of the counterweight.
(15) In some embodiments, an intended oil fill level of the crankcase chamber is between 2.5 quarts to 3 quarts.
(16) In some embodiments, the oil filter assembly includes a fill inlet so that oil can be added to the oil chamber through the oil filter.
(17) In some embodiments, wherein the filter includes filter media and an oil fill conduit extending through the filter media so that oil can be added to the oil chamber through the oil fill conduit without being filtered by the filter media.
(18) In some embodiments, the oil filter assembly includes a vent assembly in fluid communication with the oil chamber, the filter chamber, and the internal combustion engine. In some embodiments, the vent assembly is configured to allow air to flow from the filter chamber and the oil chamber to the internal combustion engine.
(19) In some embodiments, the engine assembly includes a second oil filter assembly including a second housing and a second filter. The second housing at least partially defines a second filter chamber. The second filter chamber is fluidly coupled to the supply pump and the oil gallery. The second filter is positioned within the second filter chamber. The second filter chamber is configured to filter the pressurized oil provided to the oil gallery.
(20) In some embodiments, the small air-cooled engine includes an engine block, an oil gallery configured to distribute oil, a crankcase cover, and a crankcase chamber defined by the engine block and the crankcase cover. In some embodiments, the dry sump lubrication system includes the external oil reservoir, a supply pump, and a return pump. The oil filter assembly includes a housing at least partially defining a filter chamber and a filter positioned within the filter chamber. The supply pump is in fluid communication with the oil chamber and is configured to provide pressurized oil to the oil gallery. The return pump is in fluid communication with the crankcase chamber and the filter chamber. The return pump is configured to draw oil from the crankcase chamber and provide pressurized oil to the filter chamber. The filter chamber is fluidly coupled to the supply pump and the oil gallery. The filter is configured to filter the pressurized oil provided to the oil gallery.
(21) In some embodiments, the small air-cooled engine includes an engine block, a blower housing, and a blower fan. The blower housing extends over a portion of the engine block and is configured to direct cooling air over the engine block. The blower fan is configured to pull air into the blower housing through an air inlet.
(22) Another embodiment of the invention relates to an engine assembly including an engine, an external oil reservoir, a supply pump, and a return pump. The engine includes an engine block, an oil gallery configured to distribute oil, a crankcase cover, and a crankcase chamber defined by the engine block and the crankcase cover. The external oil reservoir includes an oil tank defining an oil chamber and an oil filter assembly including a housing at least partially defining a filter chamber, wherein the filter chamber is in fluid communication with the oil chamber, and a filter positioned within the filter chamber. The supply pump is in fluid communication with the oil chamber and the crankcase chamber and the supply pump is configured to draw oil from the oil chamber and provide pressurized oil to the oil gallery. The return pump in fluid communication with the

crankcase chamber and the filter chamber and the return pump is configured to draw oil from the crankcase chamber and provide pressurized oil to the filter chamber. The filter is configured to filter the pressurized oil provided to the filter chamber. In some embodiments, the engine assembly is a component of a lawn mower. In different embodiments, the lawn mower is a riding lawn mower, a wide-area walk-behind lawn mower, a zero-turn radius lawn mower, or a standing lawn mower.

(23) Another embodiment of the invention relates to an engine assembly including a small air-cooled engine and an external oil reservoir. The small air-cooled engine includes an engine block, a crankcase cover, and a crankcase chamber defined by the engine block and the crankcase cover. The external oil reservoir includes an oil tank defining an oil chamber. The crankcase chamber and the oil chamber are sized to provide at least five hundred hours of engine oil life.

(24) Another embodiment of the invention relates to an engine assembly including a small air-cooled engine and a dry sump lubrication system. In some embodiments, the dry sump lubrication system includes an external oil reservoir and has an overall oil capacity that provides at least five hundred hours of engine oil life. In some embodiments, the engine assembly is a component of a lawn mower. In different embodiments, the lawn mower is a riding lawn mower, a wide-area walk-behind lawn mower, a zero-turn radius lawn mower, or a standing lawn mower.

(25) Another embodiment of the invention relates to an engine assembly including a small air-cooled engine and an external oil reservoir. The small air-cooled engine includes an engine block, a crankcase cover, and a crankcase chamber defined by the engine block and the crankcase cover, the crankcase chamber having a crankcase volume. The external oil reservoir includes an oil tank defining an oil chamber, the oil chamber having a reservoir volume. The reservoir volume is greater than the crankcase volume. In some embodiments, the crankcase volume and the reservoir volume have an overall oil capacity that provides at least 500 hours of engine oil life. In some embodiments, the reservoir volume is at least five quarts. In some embodiments, the reservoir volume is greater than five quarts and a recommended oil fill capacity of the oil chamber is five quarts. In some embodiments, the engine assembly is a component of a lawn mower. In different embodiments, the lawn mower is a riding lawn mower, a wide-area walk-behind lawn mower, a zero-turn radius lawn mower, or a standing lawn mower.

(26) Another embodiment of the invention relates to an external oil reservoir for use with an internal combustion engine including an oil tank defining an oil chamber and an oil filter assembly. The oil filter assembly includes a housing at least partially defining a filter chamber, wherein the filter chamber is in fluid communication with the oil chamber, a filter positioned within the filter chamber, and a return inlet in fluid communication with the filter chamber to provide oil returned from the internal combustion engine to the filter chamber to be filtered by the filter. In some embodiments, the external oil reservoir is a component of a lawn mower. In different embodiments, the lawn mower is a riding lawn mower, a wide-area walk-behind lawn mower, a zero-turn radius lawn mower, or a standing lawn mower.

(27) Another embodiment of the invention relates to an external oil reservoir for use with an internal combustion engine including an oil tank defining an oil chamber and an oil filter assembly including a filter and a return inlet configured to provide oil returned from the internal combustion engine to the filter for filtration. In some embodiments, the external oil reservoir is a component of a lawn mower. In different embodiments, the lawn mower is a riding lawn mower, a wide-area walk-behind lawn mower, a zero-turn radius lawn mower, or a standing lawn mower.

(28) Another embodiment of the invention relates to an engine assembly including an internal combustion engine and an oil tank. The engine includes at least one cylinder, a crankcase chamber, a vertical crankshaft extending through the crankcase chamber, the crankshaft configured to rotate about a vertical crankshaft axis, a supply pump including a supply inlet and a supply outlet, an oil gallery in fluid communication with the supply outlet, the oil gallery configured to distribute oil within the engine, and a return pump including an return inlet and a return outlet, wherein the

return inlet is in fluid communication with a front portion of the crankcase chamber and wherein the return pump is configured to draw oil from the front portion of the crankcase chamber through the return inlet into the return pump. The oil tank defines an oil chamber and includes a filter positioned within a filter chamber. The return outlet is in fluid communication with the filter chamber and the return pump is configured to pump oil from the front portion of the crankcase chamber to the filter chamber to be filtered by the filter. The filter chamber is in fluid communication with the oil chamber so that filtered oil passes from the filter to the oil chamber. The supply inlet is in fluid communication with the oil chamber and the supply pump is configured to pump oil from the oil chamber to the oil gallery. In some embodiments, the return pump is configured to pump oil to the filter chamber under a first pressure and pressurize the oil chamber and the supply pump is configured to draw oil from the oil chamber under a vacuum and provide oil to the oil gallery under a second pressure. In some embodiments, the first pressure is less than 15 pounds per square inch (1.034e+005 newtons/square meter). In some embodiments, the second pressure is greater than 30 pounds per square inch (2.068e+005 newtons/square meter). In some embodiments, the return pump pumps oil at a lower output pressure than the supply pump. In some embodiments, the engine assembly is a component of a lawn mower. In different embodiments, the lawn mower is a riding lawn mower, a wide-area walk-behind lawn mower, a zero-turn radius lawn mower, or a standing lawn mower.
(29) Another embodiment of the invention relates to an internal combustion engine for use with an external oil reservoir. The engine includes at least one cylinder, a crankcase chamber, a vertical crankshaft extending through the crankcase chamber, the crankshaft configured to rotate about a vertical crankshaft axis, a supply pump including a supply inlet and a supply outlet, the supply pump configured to receive oil from the external oil reservoir via the supply inlet, an oil gallery in fluid communication with the supply outlet to receive oil from the supply pump, the oil gallery configured to distribute oil within the engine, and a return pump including an return inlet and a return outlet, wherein the return inlet is in fluid communication with a front portion of the crankcase chamber and wherein the return pump is configured to draw oil from the front portion of the crankcase chamber through the return inlet into the return pump. In some embodiments, the engine is a component of a lawn mower. In different embodiments, the lawn mower is a riding lawn mower, a wide-area walk-behind lawn mower, a zero-turn radius lawn mower, or a standing lawn mower.
(30) Another embodiment of the invention relates to an oil filter for use with an oil reservoir. The oil filter includes a top including a fill inlet comprising a screen having a plurality of openings, a bottom including a filter outlet, an oil fill conduit extending between the top and the bottom and in fluid communication with the fill inlet and the filter outlet, and filter media surrounding the oil fill conduit and positioned between the top and the bottom. The filter is configured so that oil passes through the openings of the screen of the fill inlet, the oil fill conduit, and the filter outlet without being filtered by the filter media. In some embodiments, the top includes a quick connect fitting configured to secure the oil filter to the oil reservoir.
(31) Another embodiment of the invention relates to riding outdoor power equipment including an internal combustion engine and an oil reservoir. The engine includes an engine block including a first cylinder and a second cylinder arranged in a V-twin configuration, a first cylinder head for the first cylinder, and a second cylinder head for the second cylinder, wherein the first cylinder head and the second cylinder head are located at a front of the engine opposite a rear of the engine. The oil reservoir is configured for storing oil and at least a portion of the oil reservoir is located between the rear of the engine and the first cylinder head. In some embodiments, at least a second portion of the oil reservoir is located between the rear of the engine and the second cylinder head. In some embodiments, the riding outdoor power equipment also includes an operator seat and the top of the oil reservoir is located below the top of the seat. In some embodiments, at least a portion of the oil reservoir is positioned between the seat and the internal combustion engine. In some

embodiments, the entire oil reservoir is positioned between the seat and the internal combustion engine. In some embodiments, the outdoor power equipment also includes a roll bar including two upwardly extending legs, wherein the oil reservoir is positioned between the two legs of the roll bar.
(32) Another embodiment of the invention relates to outdoor power equipment including an internal combustion engine and an oil reservoir. The engine includes an engine block including a first cylinder and a second cylinder arranged in a V-twin configuration, a first cylinder head for the first cylinder, and a second cylinder head for the second cylinder, wherein the first cylinder head and the second cylinder head are located at a front of the engine opposite a rear of the engine. The oil reservoir is configured for storing oil and at least a portion of the oil reservoir is located between the rear of the engine and the first cylinder head. In some embodiments, at least a second portion of the oil reservoir is located between the rear of the engine and the second cylinder head. In some embodiments, the outdoor power equipment is a lawn mower. In different embodiments, the lawn mower is a riding lawn mower, a wide-area walk-behind lawn mower, a zero-turn radius lawn mower, or a standing lawn mower.
(33) Alternative exemplary embodiments relate to other features and combinations of features as may be generally recited in the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The invention will become more fully understood from the following detailed description, taken in conjunction with the accompanying drawings.
(2) FIG. **1** is a top view of an engine assembly including an engine and an external oil reservoir, according to an exemplary embodiment.
(3) FIG. **2** is a side view of the engine assembly of FIG. **1**.
(4) FIG. **3** is an exploded view of the crankcase cover of the engine of FIG. **1**.
(5) FIG. **4** is a top view of the crankcase cover of FIG. **3**.
(6) FIG. **5** is a perspective view of the oil reservoir of FIG. **1**.
(7) FIG. **6** is an exploded view of a portion of the oil reservoir of FIG. **5**.
(8) FIG. **6**A is a front view of an upper portion of a filter housing of the oil reservoir of FIG. **5**.
(9) FIG. **7** is a section view of the oil reservoir of FIG. **5**.
(10) FIG. **8** is a section view of an oil filter assembly, according to an exemplary embodiment.
(11) FIG. **8**A is a perspective view of the filter of the oil filter assembly of FIG. **8**.
(12) FIG. **8**B is another perspective view of the filter of the oil filter assembly of FIG. **8**.
(13) FIG. **8**C is a top view of the filter of the oil filter assembly of FIG. **8**.
(14) FIG. **8**D is a perspective view from above of a portion of the oil filter assembly of FIG. **8**.
(15) FIG. **8**E is a perspective view from below of the upper portion of the filter housing of the oil reservoir of FIG. **5**
(16) FIG. **9** is a perspective view of an external oil reservoir, according to an exemplary embodiment.
(17) FIG. **10** is an exploded view of a portion of the oil reservoir of FIG. **9**.
(18) FIG. **11** is a perspective view of a portion of the engine of FIG. **1**.
(19) FIG. **12** is a section view of a check valve, according to an exemplary embodiment.
(20) FIG. **13** is a perspective view of a portion of a lawn mower including the engine assembly of FIG. **1**.
(21) FIG. **14** is a top view of a portion of the lawn mower of FIG. **13**.
(22) FIG. **15**A is a schematic illustration of the engine assembly of FIG. **1**.
(23) FIG. **15**B is a schematic illustration of an engine assembly, according to an exemplary

embodiment.
(24) FIG. **15**C is a schematic illustration of an engine assembly, according to an exemplary embodiment.
(25) FIG. **16** is a top view of an engine assembly including an engine and an external oil reservoir, according to an exemplary embodiment.
(26) FIG. **17** is a perspective view of the engine assembly of FIG. **16**.
(27) FIG. **18** is an exploded view of the crankcase cover of the engine of FIG. **16**.
(28) FIG. **19** is a top view of the crankcase cover of FIG. **18**.
(29) FIG. **20** is a perspective view of the oil reservoir of FIG. **16**.
(30) FIG. **21** is a perspective view of a portion of a lawn mower including the engine assembly of FIG. **16**.
(31) FIG. **22** is a top view of a portion of the lawn mower of FIG. **21**.

DETAILED DESCRIPTION

(32) Before turning to the figures, which illustrate the exemplary embodiments in detail, it should be understood that the application is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology is for the purpose of description only and should not be regarded as limiting.
(33) Referring to FIGS. **1-2**, an engine assembly including an internal combustion engine **100** and an external oil tank or reservoir **112** is illustrated according to an exemplary embodiment. The internal combustion engine **100** includes an engine block **101** having two cylinders **102** and **103**, two cylinder heads **137** and **139**, two pistons, and a crankshaft **104**. Each piston reciprocates in a cylinder along a cylinder axis to drive the crankshaft **104**. The crankshaft **104** rotates about a crankshaft axis **105**. The crankshaft **104** is positioned in part within a crankcase chamber **130** defined by the engine block **101** and a sump or crankcase cover **106**. The engine **100** also includes a fuel system for supplying an air-fuel mixture to the cylinder (e.g., a carburetor, an electronic fuel injection system, a fuel direct injection system, etc.), an air filter assembly **107**, a camshaft **119** for actuating intake and exhaust valves in the cylinder heads, a muffler **108**, a flywheel, and a blower fan. The engine **100** includes a blower housing **110** configured to direct cooling air over the engine block **101** and other components of the engine. The blower fan pulls air into the blower housing **110** through an air inlet. The crankshaft **104** may be oriented horizontally (i.e., a horizontal engine) or vertically (i.e., a vertical engine). The engine may include one cylinder or two or more cylinders. The illustrated engine **100** is a vertically-shafted two cylinder engine arranged in a V-twin configuration.
(34) The engine assembly may be used in outdoor power equipment, standby generators, portable jobsite equipment, or other appropriate uses. Outdoor power equipment includes lawn mowers, riding tractors, snow throwers, pressure washers, portable generators, tillers, log splitters, zero-turn radius mowers, walk-behind mowers, wide-area walk-behind mowers, riding mowers, standing mowers, industrial vehicles such as forklifts, utility vehicles, etc. Outdoor power equipment may, for example, use an internal combustion engine to drive an implement, such as a rotary blade of a lawn mower, a pump of a pressure washer, an auger of a snow thrower, the alternator of a generator, and/or a drivetrain of the outdoor power equipment. Portable jobsite equipment includes portable light towers, mobile industrial heaters, and portable light stands.
(35) Referring to FIGS. **2-4**, the engine **100** includes a dry sump lubrication system that includes the external oil tank **112** (FIG. **2**), a supply pump **113** (FIG. **3**) for supplying oil (lubricant) from the tank **112** to the engine via oil galleries for distribution to various locations within the engine **100**, a return or scavenge pump **114** (FIG. **4**) for returning (scavenging) oil from the crankcase chamber **130** to the tank **112**, and the conduits (hoses, tubes, other plumbing) for fluidly connecting the tank **112**, the pumps **113**, **114**, the oil galleries **124**, and the crankcase chamber **130** to one another.
(36) The oil tank **112** is sized to hold oil sufficient to extend the engine oil life for a commercial lawn mower (e.g., a zero-turn lawn mower, standing mowers, etc.) to 500 plus hours. Typical

commercial lawn mower best practice is to assume 100 hours of engine oil life for a conventional engine without an external oil tank and routinely change the engine oil after 100 hours of engine operation. Providing at least 500 hours of engine oil life with the dry sump system described herein enables operation of the lawn mower (e.g., operated by a commercial lawn services provider) for an entire summer mowing season without having to change the oil, thereby reducing downtime and maintenance during the season in which the mower will be operated. The dry sump lubrication system described herein can reduce oil change labor and replacement filter costs by at least 60%. In exemplary embodiments of the dry sump system described herein about 5-6 quarts (about 4.732-5.678 liters) of oil is sufficient to provide at least 500 hours of engine oil life. In other embodiments, more or less oil may be sufficient to provide at least 500 hours of engine oil life depending on the particular arrangement of the engine used with the dry sump system. The dry sump system extends engine oil life as compared to an engine that does not include an external oil tank.

(37) Engine oil life is a measure of the operation time of an engine between recommended oil changes. During operation of an engine, the oil used to lubricate and cool the engine breaks down. Engine oil break down is due in part to heat and mechanical shearing of the hydrocarbons that make up the oil. Also, the oil may accumulate particles (e.g., metal particles worn off engine components) that negatively impact the oil's viscosity. Finally, oil contains a number of additives called an additive package, including for example anti-foaming, detergent, viscosity modifiers, and anti-corrosion additives, that help the oil perform to manufacturer and industry standards (e.g., American Petroleum Institute or Society of Automotive Engineer (“SAE”) standards). These additives are basic in nature and can be identified by the total base number (“TBN”). TBN is used as a measure of reserve alkalinity in the oil. As the engine operates the total acid number (“TAN”) increases from a value of zero and the TBN decreases. TAN is used as a measure of acid concentration in the oil. As the engine operates, the additive package loses effectiveness.

(38) As the engine oil breaks down, its viscosity increases so that the oil may appear thick, sludgy, or dirty in comparison to new oil. Manufacturers (e.g., engine manufacturers and equipment manufacturers) determine the engine oil life or recommended oil change intervals for particular lubrication systems. Signs that engine oil life for a particular lubrication system has been exceeded and an oil change should occur are when the TBN and TAN begin to approach one another (e.g., within 5%-10% of the same value) or when the TAN is greater than the TBN. Oxidation of the oil can also be measured as a way of determining engine oil life for a particular lubrication system. The viscosity of oil in a test engine can be measured as a way of determining engine oil life for a particular lubrication engine. For example, new oil measured at a rated viscosity of SAE **20** may be measured at a rated viscosity of SAE **50** after a period of operation in a test engine. This change in viscosity can be used to establish the engine oil life. Visual inspection of a test engine can also be used as a way of determining engine oil life for a particular lubrication system by evaluating the components of a test engine for signs of wear (e.g., worn bearings), by evaluating the appearance of the oil itself (e.g., dirty or dark oil), measuring the presence of wear metals in the oil (e.g., aluminum, iron, nickel, silicon, etc).

(39) FIGS. **3-4** illustrate the sump or crankcase cover **106** according to an exemplary embodiment. The pumps **113** and **114** are incorporated into the crankcase cover **106**. As illustrated, the pumps **113** and **114** are gerotor pumps. The crankcase cover **106** includes a crankshaft opening or aperture **115** through which the crankshaft **104** extends to drive one or more components of a lawn mower or other piece of equipment. A transmission **116** including one or more gears **117** or other reduction mechanism (e.g., belts) connects the crankshaft **104** to a drive shaft **118** for the supply pump **113** so the supply pump **113** is driven by the crankshaft **104** at a lower rotational speed than that of the crankshaft **104**. The supply pump **113** and the drive shaft **118** rotate about an axis of rotation **133**. The camshaft **119** is directly connected to a drive shaft **120** for the return pump **114** so the return pump **114** is directly driven by the camshaft **119** at the same rotational speed as the camshaft **119**.

The return pump **114** and the drive shaft **120** rotate about an axis of rotation **134**. A transmission connects the camshaft **119** to the crankshaft **104** so that the camshaft **119** is driven at a rotational speed less than that that of the crankshaft **104**.

(40) The supply pump **113** and the return pump **114** are located near opposite sides of the crankcase cover **106**. The supply pump **113** is located near the rear **136** of the crankcase cover **106** and the engine **100** and the return pump **114** is located near the front **138** of the crankcase cover **106** and the engine **100**.

(41) The supply pump **113** is positioned in a supply pump housing **121** formed in the exterior of the crankcase cover **106**. The supply pump **113** is in fluid communication with a supply inlet **122** for receiving oil (e.g., supply oil) from the external oil tank **112** and a supply outlet **123** for providing oil for distribution within the engine **100** by one or more oil galleries **124**. As illustrated, the supply inlet **122** includes a fitting for connecting a hose or other conduit to the supply pump **113**. The supply pump **113** pumps oil to the oil gallery **124** to distribute oil within the engine. A cover plate **125** is secured to the crankcase cover **106** (e.g., by threaded fasteners) to close the pump housing **121**. A gasket **126** is provided between the cover plate **125** and the crankcase cover **106** to form a seal.

(42) The return pump **114** is positioned in a return pump housing **127** formed in the exterior of the crankcase cover **106**. The return pump **114** is in fluid communication with a return inlet **128** for receiving oil (e.g., return oil, scavenged oil) from the crankcase chamber **130** and a return outlet **129** for providing oil to the external oil tank **112**. As illustrated, the return outlet **129** includes a fitting for connecting a hose or other conduit to the return pump **114**. A cover plate **144** is secured to the crankcase cover **106** (e.g., by threaded fasteners) to close the pump housing **127**. A gasket **146** is provided between the cover plate **144** and the crankcase cover **106** to form a seal.

(43) As shown in FIG. **4**, the return inlet **128** is formed through the crankcase cover **106** to place the return pump **114** in fluid communication with the crankcase chamber **130**. The return inlet **128** is formed in a front portion **131** of the crankcase cover **106** to be in fluid communication with the front portion of the crankcase chamber **130**. The front portion **131** is located forward of a vertical plane **132** including the vertical crankshaft axis **105**. The plane **132** is perpendicular to a second vertical plane **142** including the crankshaft axis **105** and the axis of rotation **134** of the return pump **114**. The cylinders **102** and **103** and cylinder heads are also located forward of the vertical plane **132**. The return pump **114** draws oil from the front portion **131** of the crankcase chamber **130** into the return pump **114**. The return pump **114** pumps the oil to the external oil tank **112**.

(44) Locating the return inlet **128** near the front **138** of the engine **100** helps to ensure that sufficient oil is returned from the crankcase chamber **130** to the external oil tank **112** by the return pump **114** to prevent the engine **100** (i.e., crankcase chamber, cylinders, cylinder heads, oil galleries, etc.) from overfilling with oil supplied by the supply pump **113** from the external oil tank **112**. When the lawn mower is tilted or tipped forward toward the front **138** (e.g., when traveling up or down a hill or incline), the oil in the crankcase chamber **130** moves toward the front portion **131** and the return inlet **128**, which will reduces possible occurrences of the return pump **114** being starved for oil. If the return inlet **128** were located near the rear of the engine, this orientation would cause the oil to move away from the return inlet and possibly starve the return pump. When the return pump **114** has an insufficient or otherwise unavailable oil supply (is starved for oil), the engine **100** could overfill with oil as the supply of oil from the supply pump **113** outpaces the removal of oil by the return pump **114**, potentially negatively impacting operation of the engine. Overfilling is a concern in part because the maximum volume of oil intended to be present in the engine **100** during normal operating conditions in the crankcase chamber (e.g., about 2.5 quarts-3 quarts (about 2.366 liters-2.839 liters)) is less than the volume of the external oil tank **112** (e.g., more than 5 quarts (4.732 liters)). When oil in excess of the intended maximum oil volume is present in the engine, the excess oil can impair operation of the engine (e.g., burning excess oil during combustion) and cause damage (e.g., damaging seals, gaskets, bearings, etc.). The

crankshaft **104** includes one or more counterweights **109**. As shown in FIGS. **15**A-**15**B, it is preferred to keep the recommended oil level **111** in the crankcase chamber **130** below the level of the lowermost portion of the counterweights **109** of the crankshaft **104**. When the oil level is at or above the level of the lowermost portion of the counterweights **109**, this excess oil can be transferred to the combustion chamber of the cylinders **102**, **103**, where it is combusted and causes unwanted smoke during operation of the engine **100**. Also the location of the return inlet **128** near the front **138** of the engine **100** helps the engine's performance when operated at an angle from the normal operating position shown in FIGS. **1** and **2**. Angled operation of an engine is dependent on the engine's ability to handle excessive amounts of oil in particular locations (e.g., in the cylinders) and to prevent starving the return inlet **128** of the return pump **114** for oil. During tests performed by the Applicant, the engine **100** was able to operate when positioned at an angle of 45 degrees forward, 45 degrees rearward, 45 degrees to the left, or 45 degrees to the right relative to the normal operating illustrated in FIGS. **1-2** for a period of two minutes to simulate expected angled operation by a user and for a period of one hour to simulate an extreme example of angled operation by a user.

(45) FIGS. **5-10** illustrate the external oil tank **112** of the engine **100** according an exemplary embodiment. The oil tank **112** includes a tank body **135** that defines an oil reservoir, volume, or chamber **148** for storing oil. The tank **112** may be formed from metal (e.g. aluminum) to provide advantageous heat transfer properties and/or contoured to maximize the surface area exposed to air to aid in cooling the oil stored in the chamber **148**. In some embodiments, an oil cooler is provided to further cool the oil.

(46) The external oil tank **112** includes an integrated oil filter assembly **150**. FIG. **6** illustrates the oil filter assembly **150** according to an exemplary embodiment. The oil filter assembly **150** includes an oil filter housing **152**, a cover or cap **154**, and a filter **156**. The filter **156** is a cylindrical cartridge including a cylinder of filter media or material **158** and two end caps **160** and **162**.

(47) The housing **152** includes an upper portion **164** and a lower portion **166**. The upper portion **164** includes a mount or flange **168** for securing the housing **152** to the top **170** of the oil tank **112**. The upper portion **164** also includes a substantially cylindrical filter body **172** extending upward from the flange **168**. The lower portion **166** is substantially cup-shaped with a bottom **174** and a cylindrical sidewall **176**. An opening or aperture **178** is formed through the bottom **174**. A cylindrical neck or collar **180** extends downward from the bottom **174** around the opening **178**. The lower portion **166** is attached to the upper portion **164** (e.g., by a threaded fastener, glue, adhesive, etc.) to form a filter chamber **181** for receiving the filter **156**.

(48) In some embodiments, as illustrated in FIGS. **5-7**, the flange **168** includes a dipstick opening or aperture **182** and a neck or support **184** surrounding the aperture **182**. A corresponding dipstick opening **186** is formed in the top **170** of the oil tank **112**. A dipstick **188** is guided by the neck **184** through the apertures **182** and **186** into the chamber **148** to measure the oil level therein. In other embodiments, as illustrated in FIGS. **8-10**, the flange **168** does not include components related to the dipstick **188** and instead the neck **184** extends upward from the top **170** of the oil tank **112** around a single aperture, the aperture **186** through the top **170**.

(49) The top **170** of the oil tank **112** includes an aperture or opening **190** for receiving the lower portion **166** of the housing **152**. The flange **168** is attached to the top **170** of the oil tank **112** (e.g., by threaded connection, welding, or other appropriate fastening technique) with a gasket **192** positioned between the flange **168** and the top **170**. The lower portion **166** of the housing **152** extends into the aperture **190** in the top **170** of the oil tank **112**.

(50) The lower end cap or bottom **162** of the filter **156** includes a flange **194** and a cylindrical neck or collar **196** extending downward from the flange **194**. A filter outlet, aperture, or opening **198** is formed through the flange **194** and the collar **196** to allow filtered oil to exit the filter **156**. An O-ring or gasket **200** is positioned within a circumferential groove **201** in the collar **196** to provide a seal between the lower end cap **162** and the bottom **174** of the lower portion **166** of the housing **152**

when the filter **156** is attached to the housing **152**.

(51) A cylindrical support or conduit **202** extends between and is coupled to the lower end cap **162** and the upper end cap **160**. The support **202** includes multiple openings **204** to allow oil filtered by the filter media **158** to pass through the support **202** to the filter outlet **198**. The filter media **158** (e.g., pleated filter paper or other appropriate filter material) also extends between and is coupled to the lower end cap **162** and the upper end cap **160**. The lower end of the filter media **158** contacts the flange **194** of the lower end cap **162** and a base **206** of the upper end cap **160**.

(52) As shown in FIGS. **8-8**D, the upper end cap or top **160** includes the base **206** and a sidewall **208** extending upward from the periphery of the base **206**. An O-ring or gasket **210** is positioned within a circumferential groove **211** in the sidewall **208** to provide a seal between the upper end cap **160** and the upper portion **164** of the housing **152**. Above the gasket **210**, quick-connect tabs, lugs, or protrusion **212** extends outward from the sidewall **208**. The protrusion **212** includes one or more recesses **209** to allow a corresponding protrusion **228** of the housing **152** to pass through the protrusion **212** of the end cap **160**. A fill inlet, screen, or grill **214** including one or more openings or apertures **216** functions as an oil inlet, allowing oil to flow through the upper end cap **160**, the support **202**, and the lower end cap **162** to the chamber **148** and as an air vent, allowing air to flow from the chamber **148** through the lower end cap **162**, the support **202**, and the upper end cap **160**. In some embodiments, as illustrated, the screen **214** is a frustoconical in shape with a series of circular walls **213** or members surrounding a central aperture **216** and separated from each other by apertures **216** decreasing in diameter as the screen **214** extends axially away from the base **206**. A series of ribs **215** connect the walls **213** to each other and the base **206**. A handle **217** extends through a midpoint of the screen **214** and includes a grasping portion located above the screen **214** to provide a grasping point for the user to manipulate the filter **156** as needed to attach and remove the filter **156** from the housing **152**.

(53) The filter body **172** of the upper portion **164** of the housing **152** includes an upper sidewall **218**. Two quick connect protrusions or tabs **220** extends outward from the upper sidewall **218**. A platform or shelf **224** extends inward from the upper sidewall **218** to a neck or collar **226**. The inner diameter of the upper sidewall **218** is greater than the inner diameter of the collar **226**, with the shelf **224** taking up the distance in between. The collar **226** extends downward from the shelf **224** and is spaced apart from a lower sidewall **232** to form an annular volume between the collar **226** and the lower sidewall **232**. Two quick-connect protrusions **228** are located above the shelf **224** and extend outward from the upper sidewall **218** to form a groove **230** between the protrusions **228** and the shelf **224** for receiving the protrusions **212** of the upper end cap **160** of the filter **156**. The shelf **224**, the protrusions **228**, the groove **230**, and the protrusions **212** interact to form a quarter-turn quick-connect connection between the upper end cap **160** of the filter **156** and the upper portion **164** of the housing **152** so that a user can quickly and easily attach the filter **156** to the housing **152** and quickly and easily un-attach the filter **156** from the housing **152**. Alternatively, other quick-connect arrangements or other types of attachments (e.g. a threaded attached) may be used to attach the filter **156** to the housing **152**. With the filter **156** attached to the housing **152**, the gasket **210** forms a seal between the upper end cap **160** of the filter **156** and the upper sidewall **218** of the filter body **172** of the upper portion **164** of the housing **152**.

(54) The cover **154** closes the upper end of the filter chamber **181** when attached to the upper portion **164** of the housing **152**. The cover **154** includes a groove **234** formed between an outer wall **236** and an inner wall **238**. An O-ring or gasket **239** is positioned in a groove in the inner wall **238** and forms a seal with the upper sidewall **218** of the upper portion **164** of the housing **152** when the cover **154** is attached to the housing **152**. The cover **154** and the protrusions **220** of the upper portion **164** of the housing **152** interact to form a quarter-turn quick-connect connection between the cover **154** and the upper portion **164** of the housing **152** so that a user can quickly and easily attach the cover **154** to the housing **152** and quickly and easily un-attach the cover **154** from the housing **152**. Alternatively, other quick-connect arrangements or other types of attachments (e.g. a

threaded attached) may be used to attach the cover **154** to the housing **152**.
(55) The cover **154** also includes a protrusion **240** that extends downward from the top of the cover **154** into the interior of the cover **154**. If the filter **156** is not properly attached to the housing **152**, the filter **156** will sit up higher within the filter chamber **181** than when properly attached and the protrusion **240** will contact the handle **217** of the filter **156**, preventing the cover **154** from being attached to the housing **152**. This provides a physical indication (the impact between the protrusion **240** and the handle **217**) and a visual indication (the cover **154** not seated properly on the housing **152**) to the user that the oil filter **156** has not been properly attached to the housing **152**. As shown in FIG. **8**, when the filter **156** is properly attached to the housing **152**, the protrusion **240** does not contact the handle **217** and the cover **154** can be properly seated and attached to the housing **152**. In some embodiments, the cover **154** is attached to the upper end cap **160** of the filter **156** so that the cover **154** and filter **156** may be separated from the housing **152** as a single integral unit.
(56) The upper portion **164** of the housing **152** also includes a return conduit **242** that receives return oil from the return pump **114**. A return inlet **244** is in fluid communication with the return outlet **129** of the return pump **114** (e.g., by a hose or other conduit). As illustrated, the return inlet **244** includes a fitting for connecting a hose or other conduit to the oil filter assembly **150**. The return conduit **242** is in fluid communication with the filter chamber **181** through a return outlet **246** to deliver return oil to the outside (dirty) side of the filter **156**. The return outlet **246** is located below the upper end cap **160** and the shelf **224** to deliver oil to the filter chamber **181** below the seal formed by the gasket **210** between the upper end cap **160** and the upper portion **164** of the housing **152**. The return outlet **246** is located between the seal formed by the gasket **210** between the upper end cap **160** and the upper portion **164** of the housing **152** and the seal formed by the gasket **200** between the lower end cap **162** and the lower portion **166** of the housing **152**. As shown in FIG. **8**E, the return outlet **246** is formed in the lower sidewall **232** so that return oil enters the annular volume between the lower sidewall **232** and the collar **226**. The direction of return oil flow is substantially tangential to the collar **226** to induce circular flow of the return oil through the annular volume between the lower sidewall **232** and the collar **226** to distribute the return oil around the circumference of the filter media **158**.
(57) The upper portion **164** of the housing **152** also includes a vent assembly **248** that provides a flow path for air from the external oil tank **112** to the engine **100**. The vent assembly **248** includes a filter conduit **250** in fluid communication with the filter chamber **181**, a tank conduit **252** in fluid communication with the oil chamber **148** through an opening or aperture **254** formed through the top **170** of the oil tank **112**, and an engine conduit **256** including a vent outlet **258**. As illustrated, the vent outlet **258** includes a fitting for connecting a hose or other conduit to the vent assembly **248**.
(58) All three conduits **250**, **252**, and **256** are in fluid communication with each other at a joint or manifold **259**. The conduit **250** includes an inlet aperture **251** in fluid communication with the filter chamber **181** above the upper end cap **160** (FIG. **8**D) so that the conduit **250** provides a flow path for air to exit the filter chamber **181**. The conduit **252** includes an inlet aperture **253** in fluid communication with oil the chamber **148** and positioned in the oil chamber **148** to provide a flow path for air to exit the oil chamber **148**. The conduit **256** includes an inlet aperture **257** in fluid communication with the manifold **249** to provide a flow path for air to travel from the oil tank **112** to the engine **100**.
(59) FIG. **11** illustrates a portion of the engine **100** including a vent inlet **260** for receiving air from the vent assembly **248** of the oil filter assembly **150**. As illustrated, the vent inlet **260** is includes a fitting for connecting a hose or other conduit to the engine **100**. The vent inlet **260** is in fluid communication with the crankcase chamber **130** via a vent conduit **262** to vent air from the oil tank **112** to the engine **100**.
(60) The upper portion **164** of the housing **152** also includes a bypass conduit **264**, a bypass valve **266** and a bypass passage **268**. The bypass conduit **264** is fluid communication with the filter

chamber **181** and the bypass passage **268**. A bypass inlet **265** is formed in the lower sidewall **232** of the upper portion **164** of the housing **152**. The bypass passage **268** is a volume in fluid communication with the chamber **148** of the oil tank **112** via the opening **254** formed through the top **170** of the oil tank **112** (FIG. **8**) and in fluid communication with the filter chamber **181** below the upper end cap **160** via the bypass inlet **265** (FIG. **8**E). The bypass valve **266** is positioned in the bypass conduit **264** and selectively opens and closes to allow fluid flow from the filter chamber **181** to the chamber **148** via the bypass passage **268**. The bypass valve **266** is normally-closed and pressure actuated so that when the pressure in the filter chamber **181** reaches a threshold pressure, the valve **266** opens and allows pressurized oil from the filter chamber **181** flow to the chamber **148** of the oil tank **112** without passing through the filter **156**. This may occur when the filter **156** is clogged or dirty and not able to filter the volume of oil attempting to pass through the filter **156** or during cold engine starting conditions, in which relatively high viscosity of the oil slows the rate at which oil passes through the filter **156**, causing the filter chamber **181** to fill with oil and exceed the threshold pressure of the bypass valve **266**. The tank conduit **252** of the vent assembly **248** extends downward from the manifold **259** through the bypass passage **268** and the opening **254** through the top **170** of the oil tank **112** into the chamber **148** of the oil tank **112**.

(61) As shown in FIG. **7**, the oil tank **112** includes a supply outlet **270**. As illustrated, the supply outlet **270** includes a fitting for connecting a hose or other conduit to the oil tank **112**. The supply outlet **270** is in fluid communication with a supply conduit **272** that extends into the chamber **148** of the oil tank **112**. The supply conduit **272** includes a supply inlet **274** located near a front **276** of the oil tank **112**. When a lawn mower or other equipment including the engine **100** an the oil tank **112** is tilted or tipped forward toward the front **276** (e.g., when traveling down a hill or incline), the oil in the chamber **148** moves toward the front **276** and the supply inlet **274**, which will reduces possible occurrences of the supply pump **113** being starved for oil from the oil tank **112** due to an insufficient or otherwise unavailable oil supply. A screen may be provided in the supply conduit **272** to filter large particulates and prevent them from reaching the engine **100**. A drain **278** is provided for draining oil from the chamber **148**.

(62) The recommended fill level **279** of the chamber **148** of the oil tank **112** is located below the level **281** of filter outlet **198** the oil filter **156**. This arrangement prevents the filter outlet **198** from being submerged in a volume of standing oil within the chamber **148**, which creates an unwanted backpressure on the oil filter **156** that prevents incoming oil from properly flowing through the filter. This arrangement also positions the bottom of the filter media **158** above the recommended fill level **279** which prevents the filter media **158** from being submerged in standing oil. In a preferred embodiment, the recommended fill level or recommended oil fill capacity **279** provides five quarts of oil for use by the engine **100** and the oil chamber **148** has six quarts of capacity below the level **281** of the filter outlet **198**.

(63) The tank body **135** of the oil tank **112** includes the top **170**, a bottom **280**, the front **276**, a rear **282**, a left side **284**, and a right side **286** that in combination define the volume of the oil chamber **148**. The oil tank **112** is relatively tall with its height (top to bottom) exceeding its width (left side to right side) and depth (front to rear). A recess **288** is formed in the rear **282** near the left side **284** to allow the oil tank **112** to be positioned closely to the engine **100**. The recess **288** allows the oil tank **112** to be positioned near the rear **136** of the engine **100** and next to the cylinder **103**. As shown in FIG. **1**, a volume **290** having a triangular cross-section is formed by a plane **292** extending along the rear **136** of the engine **100** (horizontal as illustrated), a plane **294** extending forward from the front outer corner of the blower housing **110** covering the cylinder **103** (vertical as illustrated and parallel to the right side **280** of the oil tank **112**), and a plane **296** extending along the outer edge of the blower housing **110** covering the cylinder **103** (angled and intersecting the planes **292** and **294** as illustrated). At least a portion of the oil tank **112** is located within the triangular cross-section of the volume **290** and is considered to be located at least in part between the rear **136** of the engine **100** and the cylinder head **139** of the cylinder **103**. In alternative

embodiments, the oil tank **112** could be similarly configured but be located between the rear **136** of the engine **100** and the cylinder head **137** of the cylinder **102**.
(64) As shown in FIG. **12**, a check valve **298** is positioned downstream of the supply inlet **122**. The check valve **298** is normally closed to prevent oil from the oil tank **112** draining to the crankcase chamber **130** through the supply conduit (e.g., when the engine **100** is not running). The check valve **298** opens and allows oil to flow from the oil tank **112** to the engine **100** when the supply pump **113** produces a vacuum in the supply port **122** sufficient to overcome the threshold of the check valve **298**.
(65) FIGS. **13** and **14** illustrate the engine **100** and the external oil tank **112** in use on a zero-turn lawn mower **300**. In other embodiments, the engine **100** and the external oil tank **112** are used with other types of outdoor power equipment, including riding lawn tractors or other riding outdoor power equipment. The engine **100** and the external oil tank **112** are located on a mounting platform **302** located between the two rear wheels **304** and **306** and behind the operator location **308**, illustrated as a seat. The engine **100** and the external oil tank **112** are also located between the vertical legs **310** and **312** of a roll bar for protecting the operator. The uppermost point of the oil tank **112** (i.e., the cover **154**) is located well below the top of the back **314** of the operator seat **308**. A portion of the oil tank **112** is located between the seat **308** and the engine **100**. Locating the tank **112** near the operator location **308**, within the vertical legs **310** and **312** of the roll bar, near the center of the mower (i.e., between the wheels **304** and **306**), and relatively low (i.e., the uppermost point below the top of the back **314** of the operator seat **308**) helps to protect the oil tank **112** from collisions (e.g., from branches or other overhanging obstacles) because the tank is close to the operator and reduces the likelihood the oil tank **112** will interfere with baggers or other accessories that may be mounted to the mower **300**.
(66) To install the oil filter **156**, the user grasps the handle **217** and inserts the oil filter **156** into the filter chamber **181** of the filter housing **152**. The collar **196** of the lower end cap **162** is inserted into the aperture **178** in the bottom **174** of the lower portion **166** of the housing **152** so that the gasket **200** forms a seal between the collar **196** of the lower end cap **162** and the collar **180** of the housing **152**. The user then rotates the filter **156** a quarter turn (i.e., 90 degrees) so that the shelf **224**, the protrusions **228**, the groove **230**, and the protrusions **212** interact to form a quarter-turn quick-connect connection between the upper end cap **160** of the filter **156** and the upper portion **164** of the housing **152** to attach the filter **156** to the housing **152**. The user may add oil to the oil tank **112** through the screen **214**. This process requires about two and a half minutes to fill the oil chamber **148** to the recommended oil tank fill level of five quarts. In some embodiments, the entire lubrication system consisting of the engine **100**, the oil tank **112**, and the conduits connecting the two has an overall recommended oil capacity or fill level of six quarts of oil. Alternatively, the user may add oil to the oil tank **112** prior to installing the filter **156** by pouring oil through the aperture **178** in the bottom **174** of the lower portion **166** of the housing **152**. This process requires less than one minute to fill the oil chamber **148** to the recommended oil tank fill level of five quarts. Filling through the screen **214** allows the user to add oil to the oil tank **112** without unseating or un-attaching the oil filter **156** from the filter housing **152**. For example, a user can top off the amount of oil in the oil tank **112** as needed after checking the fill level with the dipstick **188**, without having to remove or unseat the oil filter **156**. This reduces the chances of operating the engine **100** with the oil filter **156** not being properly attached by reducing the number of times a user needs to remove or adjust the oil filter **156**. The screen **214** prevents large debris (e.g., twigs, grass clippings, etc.) from entering the oil chamber **148** through the oil filter **156**.
(67) The cover **154** is attached by the user grasping the cover **154**, positioning the cover **154** over the protrusions **220** of the upper portion **164** of the housing **152** and rotates the cover **154** a quarter turn (i.e., 90 degrees) so that cover **154** and the protrusions **220** interact to form a quarter-turn quick-connect connection between the cover **154** and the upper portion **164** of the housing **152** to attach the cover **154** to the housing **152**. To change the oil filter **156**, the user removes the cover

154 from the housing **152**. The filter **156** is lifted upward away from the housing **152** so that excess oil will tend to drain down off of the filter **156** into the filter chamber **181** of the housing **152** and from the filter chamber **181** to the oil tank **112**. This arrangement helps to reduce the mess of dirty oil that is common when changing the oil filter of a conventional oil filter assembly.
(68) FIG. **15**A provides a schematic illustration of an engine assembly including the engine **100** and the external oil tank **112**. In operation, the return pump **114** creates a vacuum to draw in oil from the crankcase chamber **130** through the return inlet **128**. The return pump **114** pressurizes the oil (e.g., to about 5 psi (about 3.447e+004 newtons/square meter)) and distributes the pressurized return oil via the return outlet **129** to the return inlet **244** of the oil filter assembly **150** through a return oil conduit **314**. The return oil is transported via a low pressure hose, line, or conduit **314** that fluidly couples the return outlet **129** with the return inlet **244**. The return oil tends to be hot and aerated (frothy). A pressure relief valve or other device may be provided to limit pressure of the return oil provided to the oil filter assembly **150** to be below a threshold pressure. In some embodiments, the return pump **114** generates between 4 and 5 pounds per square inch (2.758e+004 and 3.447e+004 newtons/square meter) (psi) in a ⅜ inch (0.9525 centimeter) diameter return oil conduit **314** during normal operation. In some embodiments, the return pump generates less than 10 pounds per square inch (6.895e+004 newtons/square meter). This pressure provides sufficient flow of return oil to the oil tank **112** during operation of the engine **100**.
(69) As shown in FIG. **8**, the return oil enters the oil filter housing **152** through the return inlet **244** and passes through the return conduit **242** to the filter chamber **181** between the seals formed by the gasket **210** of the upper end cap **160** and the gasket **200** of the lower end cap **162**. This sealed portion of the filter chamber **181** is pressurized by the incoming return oil. The pressure forces the oil through the filter media **158**, thereby filtering particulates from the return oil and also separating air from the aerated return oil. [0085] The separated air is returned to the engine **100** via the vent assembly **248**. The air travels upward through the support **202**, passes through the screen **214** and is drawn into the filter conduit **250** of the vent assembly **248** and travels to the manifold **259**. The vent assembly **248** also allows air from the oil chamber **148** of the oil tank **112** to vent to the engine **100**. Air from the oil chamber **148** enters tank conduit **252**, mixes with air from the filter conduit **250** at the manifold **249**, and the mixture passes through the manifold **259** to the engine conduit **256**. The air leaves the engine conduit **256** through the vent outlet **258** into a vent hose, line, or conduit **316** that fluidly couples the vent outlet of the vent assembly **248** with the vent inlet **260** of the engine **100**. The vent inlet **260** is in fluid communication with the crankcase chamber **130** so that the air passes from the vent inlet **260** to the crankcase chamber. Differences in pressure between the filter chamber **181**, the oil chamber **148**, and the crankcase chamber **130** cause the air to flow as described.
(70) The filtered oil passes downward through the support **202** of the oil filter **156** and exits the filter **156** through the filter outlet **198**. From the filter outlet **198**, the filtered oil enters the chamber **148** of the oil tank **112**. There, the oil collects until being drawn back to the engine **100** by the supply pump **113**. The supply pump **113** creates a vacuum to draw oil into the supply conduit **272** through the supply inlet **274**. The oil exits the supply conduit **272** through the supply outlet **270**, which is fluidly coupled to the supply inlet **122** of the supply pump **113** by a supply oil hose, line, or conduit **318**. The oil enters the engine **100** through the supply inlet **122** and the supply pump **113** pressurizes the oil (e.g., greater than 30 psi (2.068e+005 newtons/square meter)) and supplies the oil via one or more oil galleries **124** to various locations within the engine, which may include the cylinder heads, the crankshaft, the camshaft, the crankcase chamber, various bearings, and other parts of the engine that require lubrication. In some embodiments, the oil is pressurized by the supply pump **113** to about 40-60 psi (4.137e+005 newtons/square meter).
(71) Conventional automotive dry sump lubrication systems provide the oil filter on the supply side of the system and filter the oil at high pressure, not the return side with filtering done at low pressure (e.g., less than 15 psi (1.034e+005 newtons/square meter)) as in the systems described

herein. In a conventional automotive dry sump lubrication system, oil is pumped at high pressure from the crankcase chamber through a first high pressure oil conduit to an oil filter and back from the oil filter at high pressure through a second high pressure oil conduit. By positioning the oil filter assembly **150** on the return side of the dry sump system and filtering at low pressure, the need for high pressure conduits is eliminated, resulting in cost savings and eliminating a location for a possible leak, blow-off, or other malfunction.

(72) In alternative embodiments, as shown in FIG. **15**B, a dry sump lubrication system has the oil filter assembly **150** positioned on the supply side of the dry sump system (fluidly coupled between the supply pump **113** and the oil gallery **124**) and not the return side (fluidly coupled between the return pump **114** and the oil tank **112**), as shown in FIG. **15**A. For example, as shown in FIG. **15**B, the oil filter assembly **150** is positioned downstream of the supply pump **113** to receive pressurized oil from the supply pump **113**. After the return pump **114** sends oil from the crankcase chamber **130** to the oil tank **112**, the oil is drawn by the supply pump **113** from the oil tank **112** and is sent from the supply pump **113** to the oil filter assembly **150**. After the pressurized oil is filtered by the oil filter **156**, the filtered pressurized oil exits the oil filter assembly **150** through an outlet **245** fluidly coupled to the oil gallery **124** for distribution in the engine **100**. The oil filter assembly **150** and the oil filter **156** used in this arrangement may be a conventional high pressure oil filter assembly and oil filter.

(73) In alternative embodiments, as shown in FIG. **15**C, a dry sump lubrication system includes two oil filter assemblies **150** with the first oil filter assembly **150** positioned on the supply side of the dry sump system (fluidly coupled between the supply pump **113** and the oil gallery **124**) and the second oil filter assembly **150** positioned on the return side (fluidly coupled between the return pump **114** and the oil tank **112**). This arrangement provides extra filtering capabilities as compared to the arrangements illustrated in FIGS. **15**A and **15**B.

(74) Referring to FIGS. **16-20**, an engine assembly including an internal combustion engine **400** and external oil tank or reservoir **412** is illustrated according to an exemplary embodiment. The engine **400** and the oil tank **412** are similar in many respects to the engine **100** and the oil tank **112** described above, with differences described in more detail below.

(75) As shown in FIGS. **16-17**, the internal combustion engine **400** is structurally similar to the engine **100**. The engine **400** includes an engine block **401** having two cylinders **402** and **403**, two cylinder heads (**439** shown in FIG. **17**), two pistons, and a crankshaft **404**. A crankcase chamber **430** is defined by the engine block **401** and a sump or crankcase cover **406**. The engine **400** also includes a fuel system for supplying an air-fuel mixture to the cylinder (e.g., a carburetor, an electronic fuel injection system, a fuel direct injection system, etc.), an air filter assembly **407**, a camshaft **419** for actuating intake and exhaust valves in the cylinder heads, a muffler **408**, a flywheel, and a blower fan. The engine **400** includes a blower housing **410** configured to direct cooling air over the engine block **401** and other components of the engine. The blower fan pulls air into the blower housing **410** through one or more air inlets **411**. The illustrated engine **400** is a vertically-shafted two cylinder engine arranged in a V-twin configuration.

(76) As shown in FIGS. **18-19**, the crankcase cover **406** is structurally similar to the crankcase cover **106**. The crankcase cover **406** includes a supply pump **413** and a return pump **414** that are incorporated into the crankcase cover **406**. The crankcase cover **406** includes a crankshaft opening or aperture **415** through which the crankshaft **404** extends to drive one or more components of a lawn mower or other piece of equipment. The crankshaft **404** rotates about a crankshaft axis **405**. A transmission **416** including one or more gears **417** or other reduction mechanism (e.g., belts) connects the crankshaft **404** to a drive shaft **418** for the supply pump **413** so the supply pump **413** is driven by the crankshaft **404** at a lower rotational speed than that of the crankshaft **404**. The supply pump **413** and the drive shaft **418** rotate about an axis of rotation **433**. The camshaft **419** is directly connected to a drive shaft **420** for the return pump **414** so the return pump **414** is directly driven by the camshaft **419** at the same rotational speed as the camshaft **419**. The return pump **414** and the

drive shaft **420** rotate about an axis of rotation **434**. A transmission connects the camshaft **419** to the crankshaft **404** so that the camshaft **419** is driven at a rotational speed less than that that of the crankshaft **404**. The supply pump **413** is located near the rear **426** of the crankcase cover **406** and the engine **400** and the return pump **414** is located near the front **438** of the crankcase cover **406** and the engine **400**.

(77) The supply pump **413** is positioned in a supply pump housing **421** formed in the exterior of the crankcase cover **406**. The supply pump **413** is in fluid communication with a supply inlet **422** for receiving oil (e.g., supply oil) from the external oil tank **412** and a supply outlet **423** for providing oil for distribution within the engine **400**. A cover plate **425** is secured to the crankcase cover **406** (e.g., by threaded fasteners) to close the pump housing **421**. A gasket **426** is provided between the cover plate **425** and the crankcase cover **406** to form a seal.

(78) The return pump **414** is positioned in a return pump housing **427** formed in the exterior of the crankcase cover **406**. The return pump **414** is in fluid communication with a return inlet **428** for receiving oil (e.g., return oil, scavenged oil) from the crankcase chamber **430** and a return outlet **429** for providing oil to the external oil tank **412**. The return inlet **428** is formed through the crankcase cover **406** to place the return pump **414** in fluid communication with the crankcase chamber **430**. The return inlet **428** is formed in a front portion **431** of the crankcase cover **406** to be in fluid communication with the front portion of the crankcase chamber **430**. The front portion **431** is located forward of a vertical plane **432** including the vertical crankshaft axis **405**. The plane **432** is perpendicular to a second vertical plane **442** including the crankshaft axis **405** and the axis of rotation **434** of the return pump **414**. The cylinders **402** and **403** and cylinder heads are also located forward of the vertical plane **432**. The supply pump axis of rotation **433** is located to the side of the plane **442**. The return pump **414** draws oil from the front portion **431** of the crankcase chamber **430** into the return pump **414**. A cover plate **444** is secured to the crankcase cover **406** (e.g., by threaded fasteners) to close the pump housing **427**. A gasket **446** is provided between the cover plate **444** and the crankcase cover **406** to form a seal.

(79) As shown in FIG. **20**, the oil tank **412** is structurally similar to oil tank **112**. The oil tank **412** includes a tank body **435** that defines an oil reservoir, volume, or chamber for storing oil. The tank body **435** of the oil tank **412** includes a top **470**, a bottom **580**, a front **576**, a rear **582**, a left side **584**, and a right side **586** that in combination define the volume of the oil chamber. In contrast to the oil tank **112**, which is relatively tall, the tank **412** is relatively wide with its width (left side to right side)) exceeding its height (top to bottom) and depth (front to rear). A recess **588** is formed in the rear **582** to allow the oil tank **412** to be positioned closely to the engine **400**. The recess **588** allows the oil tank **412** to be positioned near the rear **436** of the engine **100**. As shown in FIG. **16**, a volume **590** having a triangular cross-section is formed by a plane **592** extending along the rear **436** of the engine **400** (horizontal as illustrated), a plane **594** extending forward from the front outer corner of the blower housing **410** covering the cylinder **403** (vertical as illustrated), and a plane **596** extending along the outer edge of the blower housing **410** covering the cylinder **403** (angled and intersecting the planes **592** and **594** as illustrated). A volume **591** having a triangular cross-section is formed by the plane **592** extending along the rear **436** of the engine **400** (horizontal as illustrated), a plane **593** extending forward from the front outer corner of the blower housing **410** covering the cylinder **402** (vertical as illustrated), and a plane **597** extending along the outer edge of the blower housing **410** covering the cylinder **402** (angled and intersecting the planes **592** and **593** as illustrated). The oil tank **412** is located at least in part within the cross-sections of the two volumes **590** and **591** and is considered to be located at least in part between the rear **436** of the engine **400** and the cylinder head of the cylinder **103** located at least in part between the rear **436** of the engine **400** and the cylinder head of the cylinder **102**. Like the oil tank **112**, the oil tank **412** includes a supply outlet **570** and a drain **578**.

(80) The external oil tank **412** includes an integrated oil filter assembly **450** substantially the same as the oil filter assembly **150**, except for a shorter dipstick **488** to account for the shorter height of

the oil tank **412**. The oil filter assembly **450** includes an oil filter housing **452**, an oil filter (not shown), and a cover **454**.
(81) FIGS. **21-22** illustrate the engine **400** and the external oil tank **412** in use on a zero-turn lawn mower **600**. In other embodiments, the engine **400** and the external oil tank **412** are used with other types of outdoor power equipment, including riding lawn tractors or other riding outdoor power equipment. The engine **400** and the external oil tank **412** are located on a mounting platform **602** located between the two rear wheels **604** and **606** and behind the operator location **608**, illustrated as a seat. The engine **400** and the external oil tank **412** are also located between the vertical legs **610** and **612** of a roll bar for protecting the operator. The uppermost point of the oil tank **412** (i.e., the cover **454**) is located well below the top of the back **614** of the operator seat **608**. A portion of the oil tank **412** is located between the seat **608** and the engine **400**. Locating the tank **412** near the operator location **608**, within the vertical legs **610** and **612** of the roll bar, near the center of the mower (i.e., between the wheels **604** and **606**), and relatively low (i.e., the uppermost point below the top of the back **614** of the operator seat **608**) helps to protect the oil tank **412** from collisions (e.g., from branches or other overhanging obstacles) because the tank is close to the operator and reduces the likelihood the oil tank **412** will interfere with baggers or other accessories that may be mounted to the mower **600**. The engines (**100** and **400**) and external oil tanks (**112** and **412**) described herein can be used on different types of lawn mowers than the zero-turn lawn mowers (**300** and **600**) described herein. For example, the engines and external oil tank can be used on a riding mower that includes a mowing deck, a seat for the operator to sit in, and one or more blades or a drivetrain for one or more wheels (e.g., a transmission) driven by the engine. As another example, the engines and external oil tanks can be used on a wide-area walk-behind walk mower that includes a mowing deck, one or more blades or a drivetrain for one or more wheels (e.g., a transmission), and a handle that allows the user to direct and control the mower while walking behind the mower. As another example, the engines an external oil tanks can be used on a standing lawn mower that includes a mowing deck, a standing platform for the operator to stand on, and one or more blades or a drivetrain for one or more wheels (e.g., a transmission) driven by the engine.
(82) The construction and arrangement of the apparatus, systems and methods as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, some elements shown as integrally formed may be constructed from multiple parts or elements, the position of elements may be reversed or otherwise varied and the nature or number of discrete elements or positions may be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

Claims

1. A lawn mower, comprising: an air-cooled engine having two or fewer cylinders; a seat; a dry sump lubrication system including an external engine oil reservoir configured to supply oil to the air-cooled engine; and a roll bar extending above a height defined by the seat and including a first vertical leg defining a first inner surface and a second vertical leg defining a second inner surface, wherein the external oil reservoir is positioned laterally between a first plane that is parallel to and intersects the first inner surface and a second plane that is parallel to and intersects the second inner surface.

2. The lawn mower of claim 1, wherein the external oil reservoir has an oil fill capacity of at least five quarts.
3. The lawn mower of claim 2, wherein the dry sump lubrication system includes a crankcase chamber; wherein an intended oil fill level of the crankcase chamber is between 2.5 quarts to 3 quarts.
4. The lawn mower of claim 1, wherein the air-cooled engine includes a crankshaft including a counterweight positioned in a crankcase chamber; and wherein a recommended oil level of the crankcase chamber during normal operation of the engine is below a lowermost portion of the counterweight.
5. The lawn mower of claim 4, wherein an intended oil fill level of the crankcase chamber is between 2.5 quarts to 3 quarts.
6. The lawn mower of claim 1, wherein the air-cooled engine comprises: an engine block; an oil gallery configured to distribute oil; a crankcase cover; and a crankcase chamber defined by the engine block and the crankcase cover; and wherein the dry sump lubrication system comprises: the external oil reservoir comprising: an oil tank defining an oil chamber; and an oil filter assembly comprising: a housing at least partially defining a filter chamber, wherein the filter chamber is in fluid communication with the oil chamber; and a filter positioned within the filter chamber; a supply pump in fluid communication with the oil chamber and the oil gallery, wherein the supply pump is configured to draw oil from the oil chamber and provide pressurized oil to the oil gallery; and a return pump in fluid communication with the crankcase chamber and the filter chamber, wherein the return pump is configured to draw oil from the crankcase chamber and provide pressurized oil to the filter chamber; wherein the filter is configured to filter the pressurized oil provided to the filter chamber.
7. The lawn mower of claim 6, wherein the crankcase chamber has a crankcase volume and the oil chamber has a reservoir volume greater than the crankcase volume.
8. The lawn mower of claim 7, wherein the reservoir volume is at least five quarts.
9. The lawn mower of claim 6, further comprising: a crankshaft including a counterweight positioned in the crankcase chamber; wherein a recommended oil level of the crankcase chamber during normal operation of the engine is below a lowermost portion of the counterweight.
10. The lawn mower of claim 6, wherein the oil filter assembly includes a fill inlet so that oil can be added to the oil chamber through the oil filter.
11. The lawn mower of claim 6, wherein the filter comprises filter media and an oil fill conduit extending through the filter media so that oil can be added to the oil chamber through the oil fill conduit without being filtered by the filter media.
12. The lawn mower of claim 6, wherein the oil filter assembly further comprises: a vent assembly in fluid communication with the oil chamber, the filter chamber, and the internal combustion engine, wherein the vent assembly is configured to allow air to flow from the filter chamber and the oil chamber to the internal combustion engine.
13. The lawn mower of claim 6, further comprising; a second oil filter assembly comprising: a second housing at least partially defining a second filter chamber, wherein the second filter chamber is fluidly coupled to the supply pump and the oil gallery; and a second filter positioned within the second filter chamber; wherein the second filter is configured to filter the pressurized oil provided to the oil gallery.
14. The lawn mower of claim 1, wherein the air-cooled engine comprises: an engine block; an oil gallery configured to distribute oil; a crankcase cover; and a crankcase chamber defined by the engine block and the crankcase cover; and wherein the dry sump lubrication system comprises: the external oil reservoir comprising: an oil tank defining an oil chamber; and an oil filter assembly comprising: a housing at least partially defining a filter chamber; and a filter positioned within the filter chamber; a supply pump in fluid communication with the oil chamber and is configured to provide pressurized oil to the oil gallery; and a return pump in fluid communication with the

crankcase chamber and the filter chamber, wherein the return pump is configured to draw oil from the crankcase chamber and provide pressurized oil to the filter chamber; wherein the filter chamber is fluidly coupled to the supply pump and the oil gallery; and wherein the filter is configured to filter the pressurized oil provided to the oil gallery.

15. The lawn mower of claim 1, wherein the air-cooled engine comprises: an engine block; a blower housing extending over a portion of the engine block and configured to direct cooling air over the engine block; and a blower fan configured to pull air into the blower housing through an air inlet.

16. The lawn mower of claim 15, wherein the two or fewer cylinders include a first cylinder arranged within a first cylinder head and a second cylinder arranged within a second cylinder head, and wherein a portion of the external oil reservoir is arranged between a rear plane defined along a rear of the air-cooled engine and the first cylinder head.

17. A lawn mower, comprising: an air-cooled engine having two or fewer cylinders and defining a crankshaft axis; a seat; a dry sump lubrication system including an external oil reservoir, wherein the external oil reservoir is positioned between the crankshaft axis and the seat; and a roll bar extending above a height defined by the seat, and including a first vertical leg and a second vertical leg, wherein the external oil reservoir is positioned between the first vertical leg and the second vertical leg, and wherein the air-cooled engine and the external oil reservoir are positioned rearward of the seat.

18. The lawn mower of claim 17, wherein the air-cooled engine comprises: an engine block; a blower housing extending over a portion of the engine block and configured to direct cooling air over the engine block; and a blower fan configured to pull air into the blower housing through an air inlet.

19. The lawn mower of claim 1, wherein the external oil reservoir is dimensioned by a first distance and a second distance greater than the first distance, wherein the first distance extends in a longitudinal direction between the air-cooled engine and the seat, and the second distance extends in a lateral direction between the first vertical leg and the second vertical leg of the roll bar.

20. The lawn mower of claim 17, wherein the external oil reservoir is dimensioned by a first distance and a second distance greater than the first distance, wherein the first distance extends in a longitudinal direction between the air-cooled engine and the seat, and the second distance extends in a lateral direction between the first vertical leg and the second vertical leg of the roll bar.